

Measures of Central Tendency: Mean, Median and Mode

Introduction

Mean, Median, and Mode are three important measures of central tendency used in Statistics. They help us identify the **central or typical value** of a dataset.

These measures can be applied to real-life data such as:

- Daily step counts from a fitness app
 - Zomato order values
 - Spotify song durations
 - Instagram post likes
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Q1. Mean of Daily Step Counts

Definition

The **Mean** is the average value of a dataset. It is calculated by adding all observations and dividing the total by the number of observations.

Formula

Mean = $\frac{\text{Sum of all observations}}{\text{Number of observations}}$

Example

Suppose the daily step counts recorded by a fitness app are:

Day	Steps
Monday	4,000
Tuesday	5,000
Wednesday	6,000

Thursday 7,000

Friday 8,000

Calculation

First, calculate the total steps:

$$4000+5000+6000+7000+8000=30000$$

Number of observations:

$$n=5$$

Therefore,

$$\text{Mean} = \frac{30000}{5} = 6000 \text{ steps}$$

Interpretation

The average number of steps recorded per day is **6,000 steps**.

Thus, the **mean represents the average daily activity level** of the user.

Q2. Median of Zomato Order Values

Definition

The **Median** is the middle value of a dataset when all observations are arranged in **ascending or descending order**.

The median divides the data into two equal parts.

Given Data

Zomato order values:

250, 400, 150, 600, 300

Step 1: Arrange the Data

The values must first be arranged in ascending order:

150, 250, 300, 400, 600

Step 2: Find the Middle Value

There are **5 observations**.

Since 5 is an odd number, the median is the middle observation.

Median=300 $\boxed{\text{Median}=300}$

Interpretation

The median Zomato order value is **₹300**.

This means that when the orders are arranged from lowest to highest, **₹300 lies at the center** of the dataset.

Important Rule for Median

When the number of observations is odd:

The middle value is the median.

Example:

10, 20, 30, 40, 50 $10, \backslash 20, \backslash 30, \backslash 40, \backslash 50$

Median = **30**

When the number of observations is even:

The median is the average of the two middle values.

Example:

10, 20, 30, 40 $10, \backslash 20, \backslash 30, \backslash 40$

The two middle values are 20 and 30.

Median= $\frac{20+30}{2}$ $\text{Median}=\frac{20+30}{2}$ Median=25 $\boxed{\text{Median}=25}$

Q3. Mode of Spotify Song Durations

Definition

The **Mode** is the value that occurs **most frequently** in a dataset.

In this example, the values represent the duration of songs in seconds.

Example Data

Suppose a user's Spotify listening history contains the following song durations:

180, 200, 180, 240, 180, 200, 180, 200, 180, 240, 180, 200

Frequency Table

Song Duration (seconds)	Frequency
180	3
200	2
240	1

The value **180 seconds** occurs three times, which is the highest frequency.

Therefore:

Mode = 180 seconds $\boxed{\text{Mode} = 180 \text{ seconds}}$

Interpretation

The most frequently played song duration in this listening history is **180 seconds**.

Therefore, **180 seconds represents the most common song duration**.

Q4. Mean, Median and Mode of Instagram Likes

Suppose we have three different datasets representing the number of likes received by Instagram posts.

We will calculate:

- Mean
- Median
- Mode

Then we will decide which measure best represents the **typical Instagram post**.

Dataset 1: Balanced Like Counts

Data

100, 110, 120, 130, 140

Mean

$\text{Mean} = \frac{100+110+120+130+140}{5} = 120$

Median

The data is already arranged in ascending order:

100, 110, 120, 130, 140

The middle value is:

$\text{Median} = 120$

Mode

Every value occurs only once.

Therefore, there is **no mode**.

Result

Measure	Value
Mean	120
Median	120
Mode	No mode

Best Measure: Mean or Median

Both mean and median give **120 likes**.

The values are relatively close to each other and there is no extreme value. Therefore, **both mean and median are good representations of a typical post**.

Dataset 2: One Viral Post

Data

100, 110, 120, 130, 1000

Here, the post with **1,000 likes** is much higher than the other posts.

Mean

$$\text{Mean} = \frac{100 + 110 + 120 + 130 + 1000}{5} = \frac{1460}{5} = 292$$

Median

The data is already arranged:

100, 110, 120, 130, 1000

Therefore:

$$\text{Median} = 120$$

Mode

Each value occurs only once.

Therefore:

$$\text{No mode}$$

Result

Measure	Value
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Mean	292
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Median	120
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Mode	No mode
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Best Measure: Median

The **median is the best measure** for this dataset.

The reason is that **1,000 likes is an extreme value (outlier)**. It increases the mean from the central range of the other observations.

The mean of 292 suggests that a typical post receives around 292 likes, but most posts actually have only **100–130 likes**.

The median remains **120**, so it gives a more realistic representation of a typical post.

Conclusion

Median is the best measure\boxed{\text{Median is the best measure}}
because it is **less affected by extreme values**.

Dataset 3: Repeated Like Counts

Data

100, 100, 100, 120, 120, 150, 150100,\ 100,\ 100,\ 120,\ 120,\ 150,\ 150

Mean

First calculate the total:

$$100+100+100+120+120+150+150=840100+100+100+120+120+150+150=840$$

Number of observations:

$$n=7n=7$$

Therefore:

$$\text{Mean}=\frac{840}{7}\text{Mean}=\frac{840}{7}\text{Mean}=120\boxed{\text{Mean}=120}$$

Median

The data is already arranged in ascending order:

100, 100, 100, 120, 120, 150, 150100,\ 100,\ 100,\ \mathbf{120},\ 120,\ 150,\ 150

There are 7 observations, so the 4th value is the middle value.

$$\text{Median}=120\boxed{\text{Median}=120}$$

Mode

Frequency table:

Likes	Frequency
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100	3
120	2
150	2

The value 100 occurs most frequently.

Therefore:

Mode=100\boxed{\text{Mode}=100}

Result

Measure	Value
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Mean	120
Median	120
Mode	100

Best Measure: Mean or Median

Both mean and median give **120 likes**.

There is no extremely large or extremely small observation that would significantly distort the mean. Therefore, **mean or median can be used to represent the typical post**.

Overall Comparison

Dataset	Mean	Median	Mode	Best Measure
Dataset 1	120	120	No mode	Mean / Median
Dataset 2	292	120	No mode	Median
Dataset 3	120	120	100	Mean / Median

Final Interpretation

The choice of the best measure depends on the nature of the data.

Dataset 1

The data is balanced and does not contain extreme values. Therefore, **mean and median both represent the typical post well.**

Dataset 2

The dataset contains an extreme value of **1,000 likes**. Therefore, the **median is more appropriate** because it is less affected by the outlier.

Dataset 3

The values are reasonably concentrated and there is no major outlier. Therefore, **mean and median are suitable measures**. The mode is also useful for identifying the most frequently occurring number of likes.